

Ontario Broader Public Sector Industrial Energy Use and Emissions Trends 2011 - 2023

Course: Foundations of Data Science Fall 2024

Group 2: Craig Vande Ligt, Craig DCosta, Collin Potter-Bonar, Cyrus Estakhri, Colin Steensma, Chinonye Ize-Iyamu

Introduction

Climate change is one of the significant challenges of our time. As outlined by the Paris Climate Agreement (Paris Agreement, 2015, p. 1), substantial emissions reductions will be required to meet the global commitment to limit warming to 1.5 degrees Celsius above pre-industrial levels and avoid the worst impacts of climate change. According to United Nations (UN) estimates, nationally determined contributions for emissions reductions, which outline the current decarbonization plans of the 196 signatories to the Paris Agreement, imply a 2.6% reduction in greenhouse gas (GHG) emissions by 2030. Progress at this speed is insufficient. To limit warming to 1.5 degrees Celsius, the UN has determined that a 43% reduction in emissions will be required by 2030. Beyond 2030, a continued focus on achieving net-zero by 2050 will be required to meet the target. (United Nations, n.d.).

Meeting this challenge requires every sector of the economy to contribute. In the Canadian context, municipalities contribute to or otherwise influence approximately 50% of the emissions generated across the country. (Sorensen, 2022). Therefore, a fulsome understanding of energy use and GHG emissions in the municipal and broader public sector context is core to understanding Canadian progress against its commitments. Annual energy use and emissions data for Ontario's broader public sector is freely available from 2011 through 2023 in the Government of Ontario's Energy Use and Greenhouse Gas Emissions for the Broader Public Sector dataset. The dataset includes mandatory reporting for all non-ministerial public agencies in Ontario, including municipalities, municipal service boards, post secondary educational institutions, public hospitals, and school boards, among others. The data is used to inform broader public sector energy conservation and demand plans for public facilities across Ontario, including, for example, schools, universities, medical facilities, emergency services, and water treatment plants. (O. Reg. 25/23).

Focusing on broader public sector energy use, GHG emissions, and GHG emissions intensity, this analysis seeks to generate insights into Ontario's broader public sector energy use and GHG emissions profile. The analysis will determine whether emissions from industrial public sector facilities, as reported in the Government of Ontario's Energy Use and Greenhouse Gas Emissions for the Broader Public Sector dataset, are more substantial and of greater volatility over the period relative to other operation sectors.

Data Preparation

Web Scraping

Web scraping was used to capture the year and Uniform Resource Locator (URL) of the datasets (2011 - 2023) (datasets: Ministry of Energy and Electrification, 2024). A dictionary containing lists of header names per year was used to capture the appropriate data per dataset (one dataset per year).

Renaming of the headers (column names) was performed for consistency across the datasets such that a concatenation could be performed. Data attributes included the Sector, Subsector, Organization, Operation, Operation Type, City, Postal Code, Property General Floor Area (GFA) in square meters, Weekly Average Hours of the building, Source Energy

Use Intensity (EUI), Greenhouse Gas (GHG) Emissions in 1000 kilograms CO₂ equivalency (CO₂e), and GHG Emissions Intensity in kilograms CO₂e per square meter. Note that the use of these attribute/column names was done by using a capitalized convention.

Data Cleaning

The concatenated dataset, as a Pandas DataFrame, had some initial cleaning performed in which mixed data type columns were converted to the appropriate data type. An example of this was Weekly Average Hours containing "Yes" instead of a float value. The two values containing the string "Yes" were set to null data using pd.NA, and the column was then converted to a float-64 data type.

Subsequent data cleaning involved translating Operation Types from French to English using a dictionary, and standardizing the naming in the Operation Type, Subsector, and Sector. Operation Type contained a fairly large number of "other" strings. To address this, replacements for Operation Type were made based on strings found in Operation. For example, *college* found in Operation was converted to *school* in Operation Type.

Postal Code data was standardized in formatting by removing any hyphens and spaces. Some postal codes were missing the last three digits or had an incorrect sequence of letters and numbers, these were left as is as they could not be further used for the purpose of joining on latitude (explained below).

Missing Values

Missing Property GFA values were imputed using GHG Emissions and GHG Emissions Intensity data with a mask for GHG Emissions Intensity being greater than zero.

Missing values for Weekly Average Hours were imputed based on the Operation Type and Year's Weekly Average Hours mean value. Subsequent rows containing null data in this column were dropped as an appropriate value could not be imputed.

Subsequent imputations were performed to fill in missing data between GHG Emissions, GHG Emissions Intensity, and Property GFA. Since these columns contained data with values of zero, only data greater than zero per column was imputed.

Subsector data was used to impute most of the Operation Sector's undecided data (refer to Engineered Variables below). Post-cleaning left 0.38 % and 0.00% (too small to compute) of the data contributing to the total Normalized GHG Emissions Intensity as undecided and non-broad public sector (non-BPS), respectively. Since the leftover undecided contribution to the total Normalized GHG Emissions Intensity was very small, and further imputation was not possible, the data was dropped. The objective of this study focused on the Broad Public Sector's Emission data. For this reason, the non-BPS data did not belong in the dataset and was also dropped.

Engineered Variables

Two new output variables were engineered, GHG Emissions Intensity Recalculated and Normalized GHG Emissions Intensity, measured in kilograms CO₂e per square meter and kilograms CO₂e per square meter per hour, respectively. GHG Emissions Intensity

Recalculated was calculated using GHG Emissions divided by Property GFA, where Property GFA was greater than zero. Normalized GHG Emissions Intensity was calculated using GHG Emissions Intensity Recalculated divided by the Weekly Average Hours, where Weekly Average Hours was greater than zero. Note that GHG Emissions Intensity Recalculated was created due to the large number of erroneous values found in GHG Emissions Intensity. GHG Emissions was considered the source of truth on which calculations were based (further discussion is in the Discussion Section). Duplicate rows were then dropped.

Information on latitude was added using a publicly-available dataset that includes geocoordinates for each zip code and postal code in the United States and Canada (Service Objects, 2020). This comma-separated value file was merged to the Ontario emissions dataset by matching to the Postal Code variable.

A dictionary was used to convert all Operation Type values to a standardized Operation Sector value. This was another engineered attribute. The seven Operation Sectors used were (1) commercial, (2) industrial, (3) medical, (4) offices, (5) residential, (6) schools, and (7) undecided.

Extreme Outliers

Test plots and a test dataset were created to better understand the dataset and test threshold cutoff values for Property GFA, Weekly Average Hours, and Normalized GHG Emissions Intensity in an attempt to reduce the number of outliers and erroneous values. This was done due to the large volume of erroneous values found in these attributes. Most of these thresholds were not used in the cleaned dataset as large swaths of data in the industrial Operation Sector would have been removed, leading to a bias of the dataset.

Large outliers were found in the medical Operation Sector for GHG Emissions, Normalized GHG Emissions Intensity, and GHG Emissions Intensity Recalculated. Since values for these outliers existed in different years, the outliers were imputed using the mean per GHG attribute of the other years.

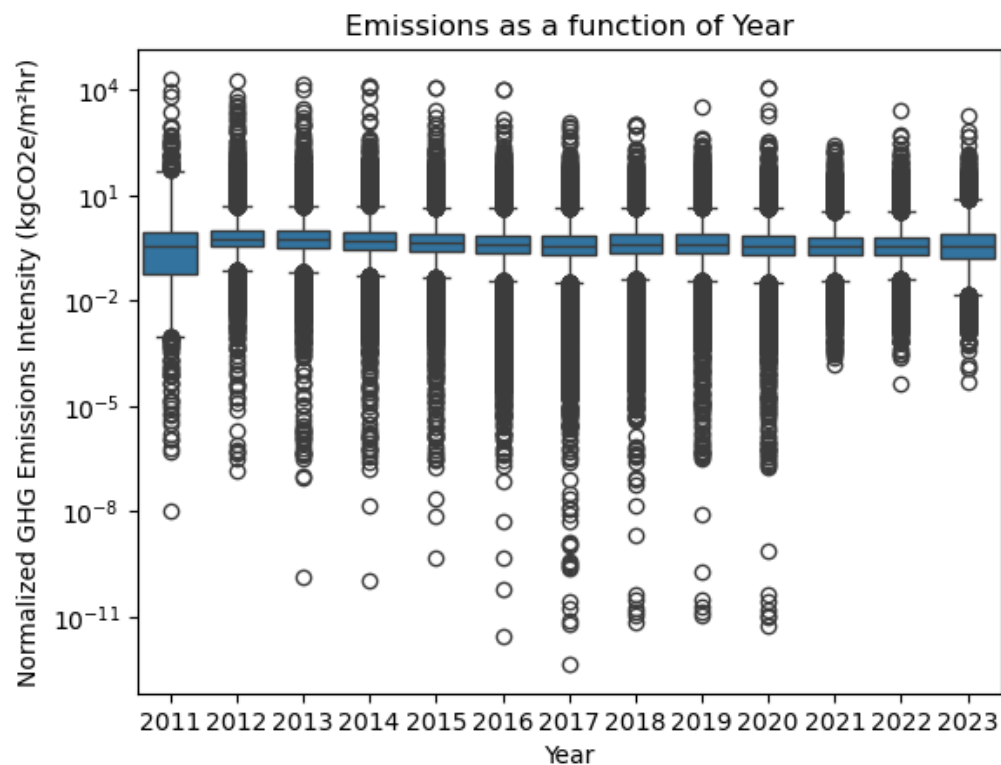
Office outliers were treated by removing duplicate data entries that caused large spikes in 2013, and imputing outliers using the mean value from other years of the same data entry.

Finally, two large outliers in 2013 and 2018 were removed from the dataset using a GHG Emissions Intensity Recalculated threshold value of less than 314,849 kg CO₂e/m².

The original dataset contained 212,271 records. After all the data preparation steps above, the dataset contained 178,555 records.

Exploration & Analysis

Overall Metrics by Year



Year	count	mean	std	min	25%	median	75%	max
2011	14449	4.46	201.10	0.00	0.06	0.35	0.87	21069
2012	15337	5.42	181.87	0.00	0.35	0.59	0.99	19467
2013	14448	4.43	153.97	0.00	0.34	0.57	1.00	14240
2014	14897	5.01	190.16	0.00	0.29	0.51	0.93	13133
2015	14885	3.59	138.84	0.00	0.26	0.44	0.82	11660
2016	15038	3.02	125.74	0.00	0.23	0.41	0.74	10777
2017	14551	1.42	17.81	0.00	0.20	0.38	0.70	1214
2018	14245	1.52	18.36	0.00	0.24	0.43	0.77	1097
2019	14367	1.77	30.44	0.00	0.23	0.41	0.77	3487
2020	14164	3.24	143.05	0.00	0.21	0.38	0.70	11813
2021	12300	0.95	5.58	0.00	0.21	0.37	0.64	285
2022	12372	1.35	26.54	-0.22	0.21	0.38	0.65	2783
2023	7502	1.77	25.95	-0.01	0.16	0.37	0.78	1973

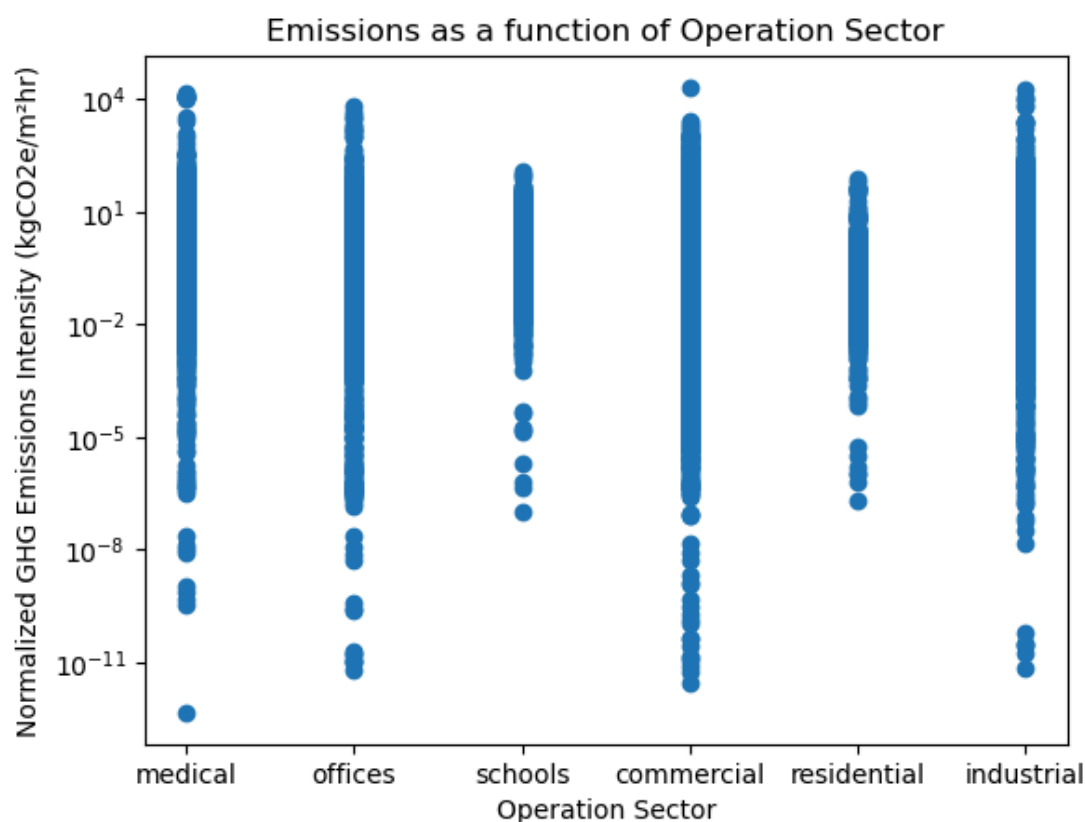
In the graph, the median of Normalized GHG Emissions Intensity for each year is represented by the horizontal line in the box of each boxplot. From this it appears that there may have been some modest reductions from 2011 until 2020 and this is reflected in the table, which shows the median generally declining each year 2012 through 2023.

In 2021 and 2022 there was a much lower range of emissions within the interquartile range (IQR) - 25% to 75% range of values represented by the upper and lower limits of the box - particularly the 75% mark.

Another feature of the Normalized GHG Emissions Intensity is the tremendous range in values, even though the values have been normalized as much as possible for the effects of area (square meters) and operating hours. This can be further noted by the need to plot the values on a log scale. According to the seaborn documentation (Seaborn Development Team, 2023), the "whiskers" coming off of the boxes in the boxplot represent values that lie within 1.5 IQRs of the upper and lower quartile. This equates to the vast majority of the values with exception of more extreme values individually plotted with circles. This large $1.5 \times \text{IQR}$ appears to be generally topping out around $100 \text{ kg CO}_2\text{e/m}^2\text{hr}$.

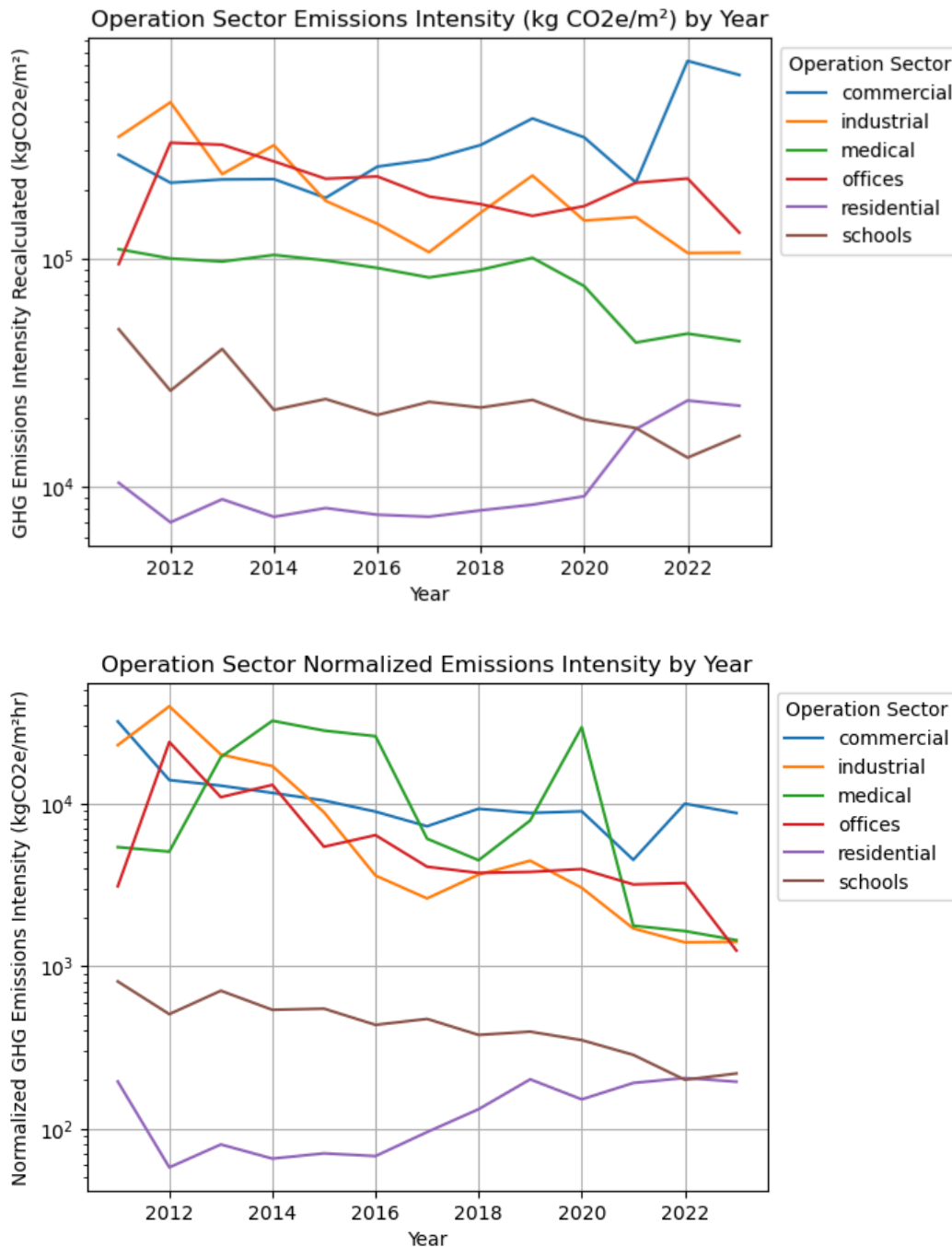
A last note on the overall metrics by year is that 2023 has a reduction in entry count of almost half compared to the prior two years, which suggests the data for that year is incomplete.

Overall Metrics by Operation Sector



Similar to the view of normalized emissions by year, each sector appears to have extreme upper and lower values, with industrial and medical having the largest standard deviation.

Operation Sector Trends Over Time

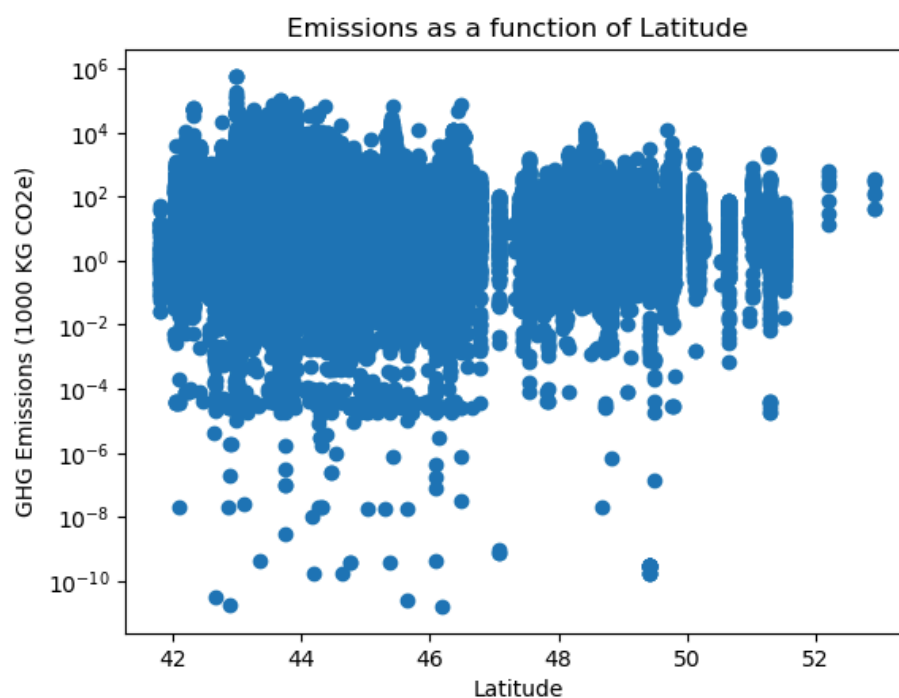


From the above graph (GHG Emissions Intensity Recalculated (kg CO₂e/m²), initial inspection shows that the hypothesis of emissions being most volatile by industrial processes is plausible. However, given the plethora of data that incorrectly provided Property GFA, the data is not fully trust-worthy and more analysis is required before finalizing the answer to the hypothesis. As expected from the pandemic, with everyone being isolated at home, a drastic increase in GHG Emissions Intensity is notable in the medical Operation Sector, and a decrease in the majority of other Operation Sectors is notable. An increase was also noted in the residential Operation Sector, as expected. What does not make sense is the small uprise in office intensity values in 2020. We'd expect to see this decrease in

2020 and 2021, not an increase. A notable spike is apparent as recovery from the pandemic in some regions began for the commercial Operation Sector in 2022.

When viewing the Normalized GHG Emissions Intensity (kg CO₂e/m²hr) data, a similar, initial highest emissions rate, is shown by the industrial Operation Sector, being taken over by both commercial and medical. Interestingly, a large spike in the 2020 medical and emergency services Operation Sector is noted. It is plausible that this is caused by the pandemic, however, further analysis is required to know for sure. The decrease seen in 2023 for the offices and commercial Operation Sector makes the accuracy of Property GFA questionable. Many locations had minimal area value, most of which did not make sense.

Latitude Analysis



A test of whether latitude correlates with total emissions was done to see whether there is a geographical weather impact. Latitude is a float value so a scatter plot and regression would occur if a trend was noted. In the scatter plot, no discernable visual trend appears, therefore latitude was not considered in the final analysis.

Statistical Analysis

To easily perform statistical analyses, a function was created to calculate all the basic statistics and quantify outliers, and then present the most useful information in summary tables. The code was rewritten a few times to narrow down the output only to the useful parts, otherwise there would be a lot of tables and graphs to read through. A second function was created specifically to assess normality with the Shapiro-Wilk test. A warning was provided that datasets greater than 5,000 entries are too large to rely on the test results, but that only applied to one operation sector, so it was ignored.

The functions were first applied to GHG Emissions Intensity Recalculated (units of $\text{kgCO}_2\text{e/m}^2$) to assess results. As the data exploration previously described, it was no surprise to find the statistics showing very high variability and no normality, but the values were quite extreme. Not a single Operation Sector in any year had a normal distribution, and the coefficients of variation (CV) were almost all greater than 100%, with many over 1,000%.

Outliers were identified as values that are more than 1.5 times the IQR above the third quartile or below the first quartile. Reviewing the outlier summary, it became obvious that they were really expanding the variability of the dataset - the industrial Operation Sector had the highest number of them, over 10% of the data entries in some years. Therefore, a function was created to copy the dataset and remove the outliers from it. Of course, with that new filtered dataset, repeating the analysis would determine new bounds and new outliers. Repeating the analysis brought the CV down below 100% for all Operation Sectors in all years except for 2011 Offices making the findings more interpretable.

The CV of industrial and school Operation Sectors both increase over time, while the others remain stable and commercial has a peak before COVID-19. While not an indicator of volatility per se, it is useful to see that variability is increasing in the industrial sector. As noted with the boxplots above, we see the mean drop in 2020 for every Operation Sector except offices, which is explainable by the pandemic. Despite dropping more than 11,000 outliers from the original dataset and reducing the outlier percentage to less than 5% for each Operation Sector & year in the cleaned dataset, there is no normal distribution of GHG Emissions Intensity Recalculated for any Operation Sector or year in the cleaned dataset.

One way to assess the initial hypothesis of volatility was to look at the mean values for each Operation Sector and year in the cleaned dataset. While the industrial Operation Sector appeared volatile with significant fluctuations from year to year, so too did the commercial Operation Sector and, to a smaller extent, the residential Operation Sector.

Not much insight was gained from the recalculated intensity variable, so the next step was to apply the functions, including the outlier remover, to the Normalized GHG Emissions Intensity variable to see if normalizing for operating hours provided greater insight. Despite dropping more than 16,000 outliers, all the insights remained the same except that the means had minimal volatility and in fact decreased over time for every Operation Sector. In general, the trend observed of the means also applied to the standard deviations.

Correlations & Relationships

The next part of the analysis was to assess the correlation between output and input variables. In the case of categorical input variables, ANOVA testing was done to assess the strength of the relationships. Interestingly, there was no correlation between the two numerical input variables (Property GFA and Weekly Average Hours) and any of the output variables apart from a very weak correlation to total emissions. Even applying the correlation function to the dataset with outliers removed did not change that result.

On the other hand, all of the categorical variables except Postal Code exhibited a very strong relationship to Normalized GHG Emissions Intensity. This was the case for every output variable, but just the one variable's results are shown in the final notebook to avoid

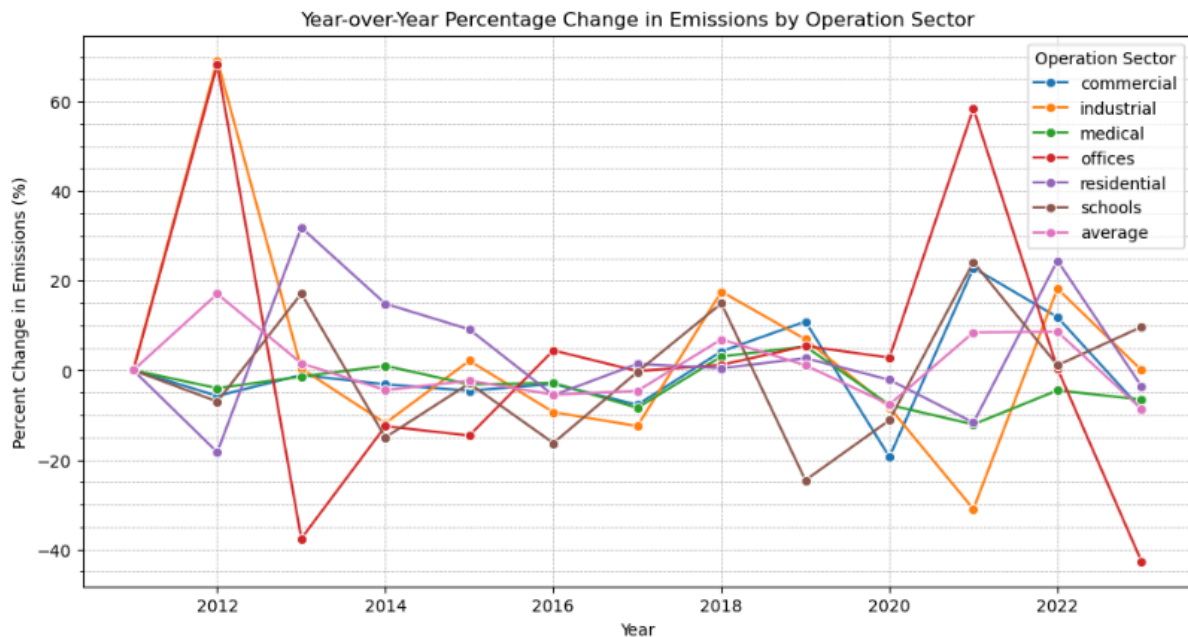
cluttering up the notebook with redundant charts and tables. The fact that Postal Code did not exhibit a strong relationship was consistent with the latitude analysis from earlier. The top 10 and bottom 10 means for each categorical variable provide some indication of what is driving (and not driving) emissions, but that does not provide enough statistical evidence to draw unbiased conclusions, rather it's guidance on where further analysis could be focused.

Further analysis of the categorical relationships could be done to find more insights, however that is more advanced statistical modelling that is out of scope for the course, and beyond the experience of the authors. Furthermore, as noted in the Discussion below, there are a lot of issues with the inputted data that could cause biased interpretation of the results.

Deltas in Emissions and Emission Intensities

Ultimately, the goal of this part of the analysis was to measure volatility in broader public sector emissions, specifically in the industrial Operation Sector. A secondary goal was determining whether emissions were on a downward or upward trajectory. GHG Emissions (1000 KG CO₂e), GHG Emissions Intensity Recalculated (kg CO₂e/m²), and Normalized GHG Emissions Intensity (kg CO₂e/m²hr) were considered in the analysis around emission reductions and volatility. To assess the variability and the extent to which emissions have decreased, emissions data and emissions intensity data were grouped according to Year and Operation Sector using the `pd.groupby` function. Once grouped, a percentage change calculation was applied to generate values that reflected the delta year-over-year and the delta from 2011 to 2023. In addition to calculating the delta for each Operations Sector year over year, the average delta across all years for each Operation Sector was also calculated.

The aforementioned analysis indicates that average emissions across the broader public sector decreased by 18.86% in 2023 from 2011. While this finding was significant and indicative of generalized progress, the year-over-year emission reduction findings demonstrated significant volatility across all Operation Sectors, notably in 2012. The most significant decarbonization occurred in the medical (-35.27%), school (-20.49%), offices (-18.21), and commercial (-9.85%) Operation Sectors, while industrial (14.13%) and residential (38.36%) Operation Sectors experienced significant increases.



The high-level analysis of Emissions Intensity Recalculated deltas suggest that emissions intensity deltas for the commercial Operation Sector had also fallen on average from 2011 to 2023. However, average intensity varies dramatically across Operation Sectors, ranging from an increase of 123.76% in the commercial Operation Sector to a 68.89% decrease in the industrial Operation Sector.

However, the Normalized GHG Emissions Intensity (kg CO₂e/m²hr) data revealed a downward trend in the mean values, standard deviations, and maximum values of this data over the years. This indicates that a significant shift in the data occurred somewhere between 2014 and 2017. The years 2011 to 2014 exhibit higher variability (indicated by larger standard deviations) compared to the later years. The lower and upper quartiles (25% and 75%) generally show a downward trend along with the mean.

It is highly likely that a fundamental change occurred in the underlying phenomenon being measured around 2017. This could be due to a range of factors, including market saturation, technological advancements, economic factors, regulatory changes, to name a few. The higher variability in earlier years could be attributed to less mature data collection processes, reporting errors, or a more dynamic environment during that period.

The results show that the commercial and industrial Operation Sectors have relatively high standard deviations, indicating greater variability in their data. This suggests that factors influencing the measured variable might have a stronger or more diverse impact in these Operation Sectors.

Linear Regression Model

The response variable of interest, greenhouse gas emissions, is expressed as a continuous variable. As such, a linear regression model was built to describe the linear relationship between a couple of the key predictor variables in the dataset and the GHG emissions response. The predictor variables were Operation Sector and Year, and the response

variable was log transformed due to the wide range of results across sectors. Two different model building processes were addressed using two different response variables.

The following was the rationale and approach for predictors:

- Operation Sector, particularly the industrial sector, was the primary predictor of interest. This variable had 6 categories which were transformed into dummy variables for independent evaluation of the effect of each sector on emissions.
- As outlined above, there was a general decline in emissions over the period covered by the data so this will likely be an important influencer on the model.

Model #1: The response variable was GHG Emissions Intensity Recalculated (kg CO₂e/m²).

In the model with both sector and year, for each increase of 1 year, there was a 1.58 increase in kilograms of CO₂e emissions per square metre of building space. In terms of Operation Sector, each sector outperformed the industrial sector as anticipated; each sector had between 45 and 89 fewer kilograms of CO₂e emissions per square metre of building space compared to industrial sector buildings.

The sector results were highly significant with p-values lower than 0.0001. The year result was not significant at the p<0.05 level and the 95% confidence interval included the null (0 value) so the association was rejected.

The adjusted r-squared value was extremely low, indicating that these variables contribute very little to explaining emissions per building space: at 0.001 it only explains 0.1% of the variance. This was likely due to the massive emissions values still skewing the model despite the authors' efforts to eliminate obvious outliers. The skew from the skewness value should be close to 0 for normally distributed data but here was 133.

Model #2: The response variable was Normalized GHG Emissions Intensity (kg CO₂e/m²hr). The models now included some control for operating hours, showing some differences compared to the previous set of models.

For the model of both sector and year, for each increase of 1 year, there was a 0.35 decrease in kilograms of CO₂e emissions per square metre of building space per operating hour. In terms of Operation Sector, almost all sectors outperformed the industrial sector as anticipated with the exception of medical; each other sector had between 3.2 and 5.2 fewer kilograms of CO₂e emissions per square metre of building space per operating hour compared to industrial sector buildings. For the medical sector there was actually 1.6 more kg CO₂e/m²hr, although the p-value was not less than 0.05 and the 95% CI included the null 0 value, so the finding was rejected.

The other Operation Sector results were significant with p-values that were lower than 0.05. The Year result was highly significant with p<0.0001. The adjusted r-squared value was still extremely low which indicates that these variables contribute very little to explaining emissions per building space. This was likely due to the massive emissions values still skewing the model despite the authors' efforts to eliminate obvious outliers. The skewness value was still high at 107.

Discussion

The analysis generated meaningful conversation among the collaborating authors, throughout the process and following it. The following represents a high-level overview of some material topics.

Data Implications on Results

GHG Emissions was considered the source of truth for calculating GHG Emissions Intensity Recalculated and Normalized GHG Emissions Intensity. Discrepancies between the recalculated values and those of GHG Emissions Intensity illustrated that significant errors were present throughout the dataset. Further investigation was performed upon which discreditable Property GFA values were found, including GFAs of zero. Average Weekly Hours also contained erroneous values.

It is believed that user input error in the form of erroneous values due to incorrectly placed decimals and assumptions about data units (improper unit conversions) may be at fault for some of the erroneous values found in the dataset. As found by the authors (Ministry of Energy and Electrification, 2024b), a 52-page user guide describing how to input the data made the task tediously long, leaving speculation open about if users read the guide in its entirety, leading to user input errors.

While GHG Emissions was the source of truth from which calculations were made, it cannot be used as the answer to our hypothesis regarding the volatility of the industrial Operation Sector being higher over the period relative to other Operation Sectors. This is because it does not account for the operation's size, nor the number of hours on average the operation is running for. Normalizing the data based on the operation size to obtain an intensity value, as done with GHG Emissions Intensity Recalculated, and by both the operation size and average weekly hours to obtain an hourly intensity value, ensuring proper comparisons could be made across Operation Sectors.

Decarbonizing the Broader Public Sector

Ontario's broader public sector has reduced its GHG emissions. This analysis indicates that emissions across the broader public sector have fallen by 18.86% in 2023 from 2011. Our findings agree with downward trends identified by the Environmental Commissioner of Ontario. The Commissioner determined that the public service in Ontario surpassed their 2021 target, which called for a 27% GHG emissions reduction in 2021 from 2006. (Saxe, 2019, p. 204).

While Ontario exceeded its 2021 target by generating a 34% reduction over the period (Saxe, 2019, p. 204), it has yet to achieve reduction rates that align with Canada's commitment of a 40% to 45% reduction in 2030 from 2005. (Environment and Climate Change Canada, 2023). Given that the linear regression model included in this analysis forecasts continued reductions in the broader public sector's GHG emissions, aligning with the Government of Canada's targets appears within reach.

There are a range of opportunities for Ontario's broader public sector to accelerate its momentum. Examples of decarbonization activities for broader public sector facilities include:

- **Comprehensive local energy planning:** While energy policy falls under the purview of provincial governments, energy operators – like Ontario's Independent Electricity System Operator – and energy distributors, local governments and the broader public sector can play a meaningful role in decarbonizing local energy while building a more resilient system by preparing comprehensive energy plans and developing community electricity systems. Local planning is already underway in some Ontario municipalities and counties, including in Hamilton, Ottawa, Oxford, and Toronto. (Government of Ontario, 2024).
- **Energy efficiency:** Broader public sector entities have a significant opportunity to improve emissions performance by enhancing the energy efficiency of their facilities. This can take many forms, including the installation of air source heat pumps, envelope and insulation upgrades as outlined in the University of Toronto Mississauga's Conservation and Demand Management Plan. (University of Toronto Mississauga, 2024, p. 6).
- **Demand management:** Across operation sectors, broader public sector entities can generate emissions savings by actively managing demand. The majority of the fossil fuel derived electricity in Ontario is generated and consumed during peak electricity consumption periods. (Saxe, 2019, p. 18). Conserving energy, fuel switching, or transitioning to renewable energy during these periods can be especially impactful. (City of Kingston, 2024, p. 18).
- **Stakeholder engagement strategies:** Engaging stakeholder groups and members of the community can have a material impact on emissions. The City of London launched its Culture of Conservation program in 2012. The program engages the municipal civil service in its conservation efforts by providing tools, resources, and policy instruments to its various departments. The City has also developed recognition mechanisms to reward departments that perform to a high standard. Similar programs can be instituted at facilities across the broader public sector, regardless of their operation type. (City of London, 2024, p. 13).

The illustrative examples of decarbonization opportunities outlined above may represent significant upfront investment on the part of Ontario's broader public sector. However, funding opportunities, like the grants distributed through the Green Municipal Fund (Federation of Canadian Municipalities, n.d.), are available to broader public sector entities in partnership with municipal governments. In addition to the funding opportunities for decarbonization projects, and perhaps more importantly, some initiatives will generate financial savings. Indeed, the Environmental Commissioner of Ontario asserts that energy conservation efforts can generate annual savings of \$450 million in utility bill savings across the broader public sector. (Saxe, 2019, p. 23).

Given the influence that the broader public sector has on overall emissions in Canada (Sorensen, 2022) and the fact that many in the private sector have started walking back their

climate commitments (Pucker, Kenneth P., 2024), it is critical that the broader public sector and the communities it serves demonstrate leadership through increased ambition.

Conclusion

The analysis concluded that there have been modest declines in greenhouse gas emissions in Ontario's broader public sector from 2011 to 2023. The analysis also points to both greater emissions output and greater volatility of emissions in the industrial operating sector as compared to other public operating sectors. However, the exploration and analysis of the data brought to light serious limitations in interpreting the data in any sort of conclusive fashion. This limitation was directly related to the extreme range in emissions values encountered. While there could be legitimate differences in real emissions outputs between Operation Sectors and across years, the investigation of the various input variables demonstrated that there is a strong possibility that data were erroneously input in numerous instances by building managers completing this survey. While best attempts were made to isolate and impute values for the most obvious erroneous entries, challenges were still encountered with this data which compromised a number of assumptions around normality of the distribution, and limited the ability to draw more definitive conclusions.

Recommendations to the government of Ontario were twofold. Firstly, while the decarbonization process does appear to be proceeding in the broader public sector, there is much room for improvement to the process of reducing emissions and, as such, a number of additional processes to be put in place to advance decarbonization activities as outlined above have been proposed. Secondly, it is recommended that the government of Ontario make significant investments in its broader public sector energy and emissions data collection process. This could include automating a number of steps to remove human error from the process. For instance, data on area, in square meters, of buildings are likely available in other government documentation detailing publicly owned real estate. As such the information could be merged into the survey data automatically. Similarly, energy use could be input directly from gas or other meter readings. Even if all buildings cannot initially conform to this automated data entry, a secondary dataset could be created with this more reliable data from a subset of public buildings to validate the results provided by the existing survey. Given the seriousness of the climate crisis for our planet's well-being, the importance of the Ontario broader public sector as a key energy consumer in one of the largest jurisdictions in North America, and the significant challenges of drawing definitive conclusions from these emissions data due to likely data entry errors, it is in the interests of not only the government of Ontario, but also the future of all of its citizens to make these changes without delay.

References

- Brander, M. (2012). *Greenhouse Gases, CO₂, CO₂e, and Carbon: What Do All These Terms Mean?* Ecometrica.
<https://ecometrica.com/assets/GHGs-CO2-CO2e-and-Carbon-What-Do-These-Mean-v2.1.pdf>
- Broader Public Sector: Energy Reporting and Conservation and Demand Management Plans, O. Reg. 25/23, Government of Ontario (2023). Retrieved from
<https://www.ontario.ca/laws/regulation/r23025?form=MG0AV3>
- City of Kingston. (2024). *Energy Conservation and Demand Management Plan*. City of Kingston.
https://www.cityofkingston.ca/media/0blpykyj/facilities_plan_energyconservationdemandmanagement.pdf
- City of London. (2024). 2024-2028 Corporate Conservation and Demand Management Plan. City of London.
https://ehq-production-canada.s3.ca-central-1.amazonaws.com/2ac986f24213c4eb15f91d08cf0c5aaa2f9502b5/original/1727361488/83b6224077130c089617747a8368c541_2024_CDM_Final.pdf?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIA4KKNQAKIOR7VAOP4%2F20241208%2Fca-central-1%2Fs3%2Faws4_request&X-Amz-Date=20241208T044533Z&X-Amz-Expires=300&X-Amz-SignedHeaders=host&X-Amz-Signature=07b444ef90ed4044fcd3734316723a7e10a1760cf4483f37bcb0f7edbbba91ae7
- Energy Star. (n.d.). *The Difference Between Source and Site Energy*. Retrieved November 14, 2024, from
<https://www.energystar.gov/buildings/benchmark/understand-metrics/source-site-difference>
- Environment and Climate Change Canada. (2024, May 2). *Global Warming Potentials*. Retrieved December 7, 2024, from
<https://www.canada.ca/en/environment-climate-change/services/climate-change/greenhouse-gas-emissions/quantification-guidance/global-warming-potentials.html>
- Environment and Climate Change Canada. (2023, December 21). *Greenhouse Gas Emissions Projections* [Research].
<https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/greenhouse-gas-emissions-projections.html>
- Federation of Canadian Municipalities. (n.d.). *Green Municipal Fund*. Retrieved December 8, 2024, from <https://greenmunicipalfund.ca/>
- Government of Ontario. (2024, January 19). Ontario's Clean Energy Opportunity: Report of the Electrification and Energy Transition Panel. Planning for Electrification and the Energy Transition.
<http://www.ontario.ca/document/ontarios-clean-energy-opportunity-report-electrification-and-energy-transition-panel-7>
- Green Municipal Fund. (n.d.). *Accelerating Community Energy Systems*. Retrieved December 7, 2024, from
<https://greenmunicipalfund.ca/funding/accelerating-community-energy-systems>

- Ministry of Energy and Electrification. (n.d.). *Broader Public Sector Energy Reporting*. Retrieved December 7, 2024, from <http://www.ontario.ca/page/broader-public-sector-energy-reporting>
- Ministry of Energy and Electrification. (2024). *Energy Use and Greenhouse Gas Emissions for the Broader Public Sector* [Dataset]. Ontario Data Catalogue. <https://data.ontario.ca/dataset/energy-use-and-greenhouse-gas-emissions-for-the-broader-public-sector>
- Ministry of Energy and Electrification. (2024a). *Energy and Water Usage of Large Buildings in Ontario* [Dataset]. Ontario Data Catalogue. <https://data.ontario.ca/dataset/energy-and-water-usage-of-large-buildings-in-ontario>
- Ministry of Energy and Electrification. (2017, May). *Fuels Technical Report*. Government of Ontario. <http://www.ontario.ca/document/fuels-technical-report/module-1-additional-information-about-fuels-and-supply>
- Ministry of Energy and Electrification. (2024b). *Guide to Submitting Data through Portfolio Manager for Ontario's Broader Public Sector*. Government of Ontario. <https://www.ontario.ca/files/2024-03/energy-broader-public-sector-bps-portfolio-manager-guide-en-2024-03-27.pdf>
- Ministry of Energy and Electrification. (2024c). *List of Building Types*. Guide to Energy and Water Reporting. Government of Ontario. <http://www.ontario.ca/document/guide-energy-and-water-reporting/list-building-types>
- Paris Agreement, United Nations (2015). https://unfccc.int/process-and-meetings/the-paris-agreement?gad_source=1&qclid=CjQKcQIAGdC6BhCgARIsAPWNWH3fgz20NGVzKUrw0u2JqpNC1V_bAPN0RwA0KaFguQbY3hlnxsSW1AaAieeEALw_wcB
- Pucker, Kenneth P. (2024, August 20). *Companies Are Scaling Back Sustainability Pledges. Here's What They Should Do Instead*. Harvard Business Review. <https://hbr.org/2024/08/companies-are-scaling-back-sustainability-pledges-heres-what-they-should-do-instead>
- Saxe, D. (2019, March). *A Healthy, Happy, Prosperous Ontario: 2019 Energy Conservation Progress Report*. Environmental Commissioner of Ontario. https://www.auditor.on.ca/en/content/reporttopics/envreports/env19/2019_EnergyConservationProgressReport.pdf
- Seaborn Development Team. (2023). *Categorical Plots - Boxplots*. Seaborn Documentation. Retrieved from <https://seaborn.pydata.org/tutorial/categorical.html#boxplots>
- Service Objects. (2020). *Download Free ZIP and Postal Code Database with GeoCoordinates for US and Canada*. Retrieved November 20, 2024 from <https://www.serviceobjects.com/blog/free-zip-code-and-postal-code-database-with-geo-coordinates/>
- Sorensen, K. (2022, March 21). *Municipalities Can Play a Lead Role in Reducing Emissions: Senator Sorensen*. Senate of Canada. Retrieved December 7, 2024, from <https://sencanada.ca/en/sencaplus/opinion/municipalities-can-play-a-lead-role-in-reducing-emissions-senator-sorensen/>

United Nations. (n.d.). *For a Livable Climate: Net-zero Commitments Must Be Backed by Credible Action*. United Nations Net Zero Coalition; United Nations. Retrieved December 7, 2024, from <https://www.un.org/en/climatechange/net-zero-coalition>

University of Toronto Mississauga. (2024). *Broader Public Sector Energy Conservation and Demand Management Plan (Updated)*. University of Toronto Mississauga. <https://www.utm.utoronto.ca/green/sites/files/green/documents/2024-08/2024%20CDM%20Plan%20-%20Defy%20Gravity%20R3.pdf>

US EPA. (2015, December 29). *Sources of Greenhouse Gas Emissions* [Overviews and Factsheets]. Retrieved December 7, 2024, from <https://www.epa.gov/ghgemissions/sources-greenhouse-gas-emissions>